

- **ANSWER THE FOLLOWING QUESTIONS**
- Figure in bracket [] next to each question indicates marks for that question
- Special Notation in the questions: **X & Y**: First and the Second last digit of your ID, respectively.

**Question No. 1**

- (a) For the circuit shown Fig.1, write the super mesh equation. [4]

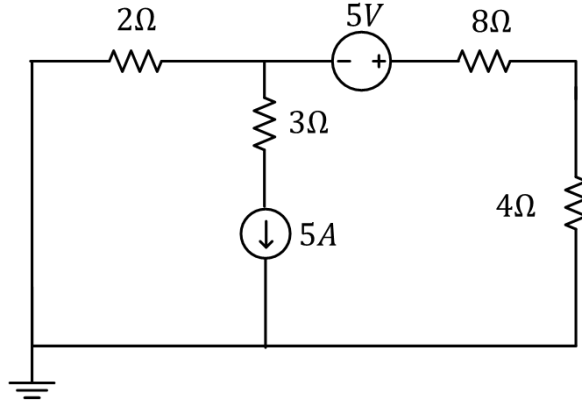


Fig.1

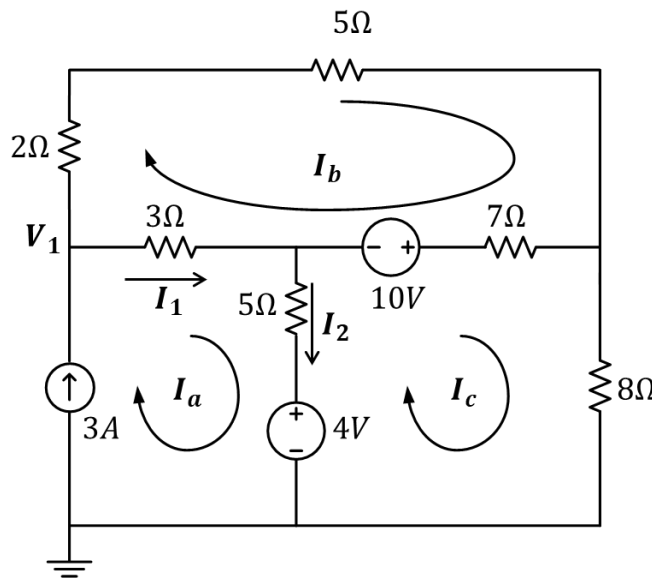


Fig.2.

- (b) For the circuit shown in Fig.2, the mesh currents are given as: . Find the branch current  $I_1$  and  $I_2$  and the terminal/node voltage  $V_1$ . [5]
- (c) Determine the time constant for the circuit shown in Fig.3.[4]

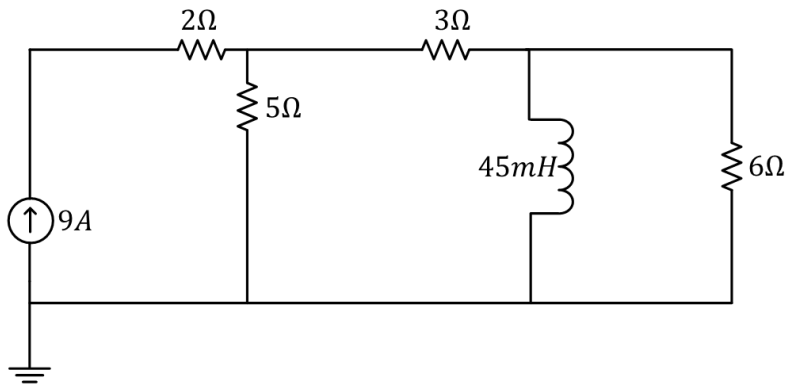


Fig.3

**Question No. 2**

- (a) Using superposition principle, determine  $V_1$  and  $I_1$  for the circuit shown in Fig.4. [10][CO2-PO-a, P1, P3]

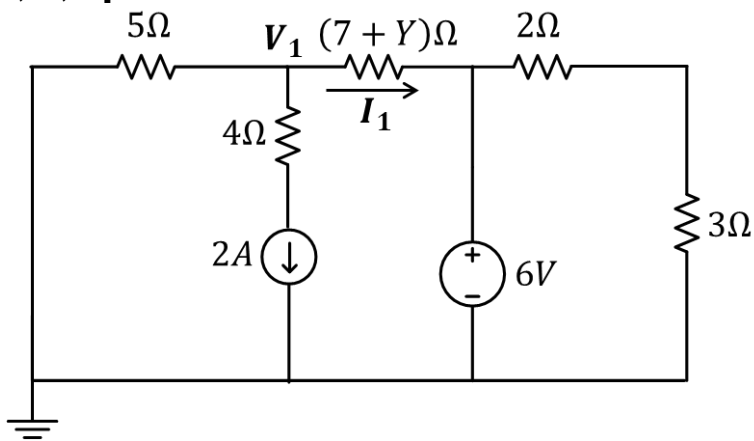


Fig.4.

- (b) Find the Thevenin equivalent circuit of the circuit shown in Fig.5 to the left of the terminals **a-b**. [15][CO2-PO-a, P1, P3]

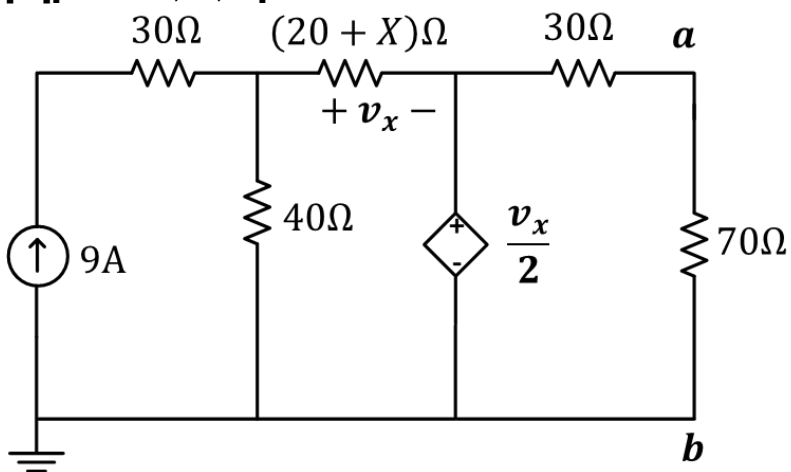
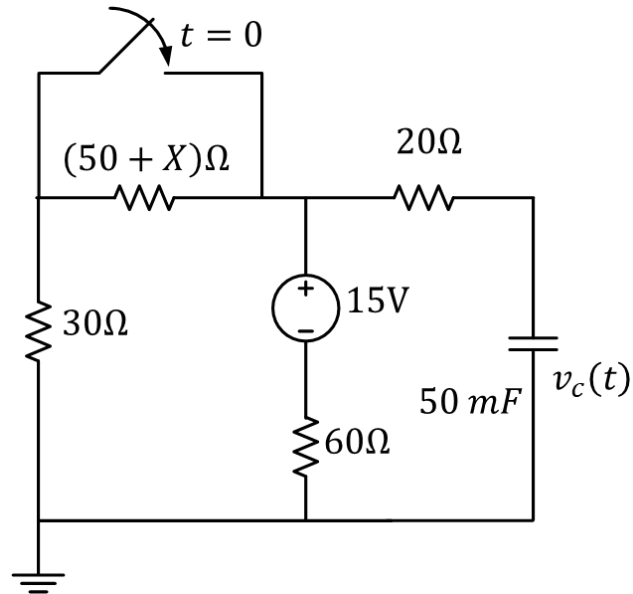


Fig.5

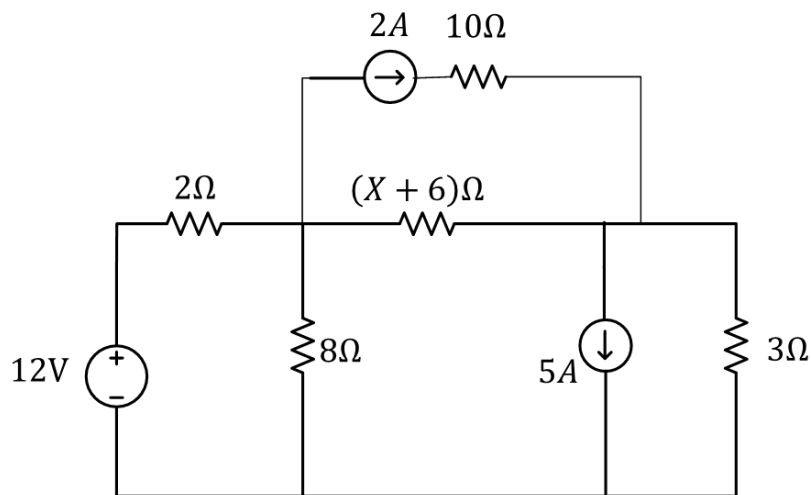
**Question No. 3**

- (a) The switch in the circuit shown in Fig.6 has been open for a long time and closed at  $t = 0$ . Determine  $v_c(t)$  for
- $t < 0$ . [4]
  - $t > 0$ . [10]



**Fig.6.**

- (b) Employing repeated source transformations, reduce the circuit such that it contains only a resistor and a voltage source in series. [8][Bonus Question]



**Fig.4**

**Fig.5.**

**THE END**